Combinatorics

Chapter 8

Definition 8.1 Let n and k be two natural numbers. Define the **Binomial Coefficient** $\binom{n}{k}$ (reads "n choose k") to be the number of subsets of exactly k elements in set of n elements. That is, let S be a set of n elements, then

$$\binom{n}{k} = |\{U \subseteq S : |U| = k\}|$$

Example Compute

- $\binom{4}{3}$. Consider the set $S = [4] = \{1, 2, 3, 4\}$. The subsets of S of three elements are given by
$$\{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}$$
Therefore, $\binom{4}{3} = 4$
- $\binom{3}{5}$. There are no subsets of 5 elements in a set of 3 elements, therefore $\binom{3}{5} = 0$
- $\binom{4}{2}$. Consider the set $S = [4] = \{1, 2, 3, 4\}$. The subsets of S of three elements are given by
$$\{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}$$
Therefore, $\binom{4}{2} = 6$

Principle (Addition Principle) If $A_0, A_1, \ldots, A_n$ is a partition for a set X, then

$$|X| = \sum_{k=0}^{n} |A_k|$$

Theorem 8.2 Let $n \in \mathbb{N}$. Recall $[n] = \{m \in \mathbb{N} : 1 \leq m \leq n\}$. Then

$$\sum_{k=0}^{n} \binom{n}{k} = |\mathcal{P}([n])|$$

Proof. For any integer k between 0 and n, define

$$A_k = \{U \in \mathcal{P}([n]) : |U| = k\}$$

Then by definition, $|A_k| = \binom{n}{k}$. We show that family $\{A_k : 0 \leq k \leq n\}$ is a partition for $\mathcal{P}([n])$.

- For $0 \leq k \leq n$, let $U = \{m \in [n] : 1 \leq m \leq k\}$. Then $U \in A_k$, therefore A_k is nonempty.
- Let $1 \leq k, l \leq n$ be such that $A_k \neq A_l$. Assume seeking a contradiction that there is a set $U \in A_k \cap A_l$. Then by definition, $|U| = k$ and $|U| = l$, which

implies that $k = l$. This contradicts the fact that $A_k \neq A_l$.

- Finally, we show that $\mathcal{P}([n]) = \bigcup_{k=0}^{n} A_k$. Since each A_k is a subset of $\mathcal{P}([n])$ by definition, then so is $\bigcup_{k=0}^{n} A_k$. Conversely, suppose that $U \in \mathcal{P}([n])$. Then $U \subset [n]$ which implies that $|U| = k$, for some $0 \leq k \leq n$. This means that $U \in A_k$ and hence $U \in \bigcup_{k=0}^{n} A_k$, as desired.

We just shows that $\{A_k : 0 \leq k \leq n\}$ is a partition for $\mathcal{P}([n])$, therefore by the Addition Principle,

$$|\mathcal{P}([n])| = \sum_{k=0}^{n} |A_k| = \sum_{k=0}^{n} \binom{n}{k}$$

$\square$

Principle (Multiplication Principle) Let X be a set satisfying the following:

- Each element of X can be described uniquely in a sequence of r-steps.
- The k^{th}-step, for $1 \leq k \leq r$, has n_k choices and can only depend on the previous choices.
- n_k, the number of choices, must be independent of the previous steps. Then, $|X| = n_1 n_2 \dots n_r$.

Example Find the cardinalities of the following sets:

1. $X = \{(a, b) : a, b \in [5]\}$
   Using the multiplication principle, (a, b) can be determined by:
   - Choose $a \in [5]$, 5 choices
   - Choose $b \in [5]$, 5 choices

   Therefore $|X| = 25$
2. $Y = \{(a, b) : a, b \in [5] \text{ and } a \neq b\}$
   Using the multiplication principle, (a, b) can be determined by:
   - Choose $a \in [5]$, 5 choices
   - Choose $b \in [5] \setminus \{a\}$, 4 choices

   Therefore $|Y| = 20$
3. $Z = \{(a, b) : a, b \in [5] \text{ and } a \leq b\}$
   Notice that not only will the choice of b depend on the choice of a, but the number of choices for b will depend on the choice of a, so we can't use the multiplication principle for Z. We do the following, define

   $$A_k = \{(k, b) : b \in [5] \text{ and } k \neq b\}, \text{ for all } k \in [5]$$

   To determine $(k, b) \in A_k$, we only have to choose $b \in [5]$ and$k \leq b$, $6 - k$ choices. Therefore $|A_k| = 6 - k$, and by the multiplication principle:

   $$|Z| = \sum_{k=1}^{5} (6 - k) = 15$$

Example Let X be a set of n elements. Consider the following list of elements $X = \{x_1, x_2, \ldots, x_n\}$. Then

$$|\mathcal{P}(X)| = 2^n$$

In order to determine $U \in \mathcal{P}(X)$ we

- Choose whether or not x_1 is an element of U, 2 choices
- Choose whether or not x_2 is an element of U, 2 choices

  $\vdots$
- Choose whether or not x_n is an element of U, 2 choices

Therefore $|\mathcal{P}(X)| = 2^n$

Principle (Double Counting Principle) order to prove a combinatorial equation, it suffices to do the following

- Define a set X with an appropriate number of elements
- Prove that $|X|$ equals to the LHS of the given equation
- Prove that $|X|$ equals to the RHS of the given equation

Definition 8.3 Let n be a natural numbers. Define the **factorial of** n to be the number of bijections from $[n]$ to $[n]$ (Equivalently, $n!$ is the number bijections $X \to Y$, wheere $|X| = |Y| = n$)

Theorem 8.4 For $n \in \mathbb{N}$,

$$n! = \prod_{k=1}^{n} k$$

Proof. We use the Double Counting Principle as follows: Let X be the set of all bijections from $[n]$ to $[n]$. Then $|X| = n!$, be definition. On the other hand, we can find $|X|$ using the Multiplication Principle as follows: In order to determine $f : [n] \to [n]$ we

- Choose $f(1) \in [n]$, n choices
- Choose $f(2) \in [n] \setminus \{f(1)\}$, $n-1$ choices

  $\vdots$
- Choose $f(k) \in [n] \setminus \{f(1), f(2), \ldots, f(k-1)\}$, $n-k$ choices

  $\vdots$
- Choose $f(n) \in [n] \setminus \{f(1), f(2), \ldots, f(n-1)\}$, 1 choice

Therefore, $|X| = n \cdot (n-1)\dot{2} \cdot 1$. $\square$

Theorem 8.5 Let $n, k \in \mathbb{N}$ with $k \leq n$, then

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Proof. We will prove that

$$\binom{n}{k} k!(n-k)! = n!$$

using double counting. Let X be the set of all bijections form $[n]$ to $[n]$. Then

$$|X| = n! \tag{1}$$

by definition. We show that $|X| = \binom{n}{k} k!(n-k)!$ by the multiplication principle. To uniquely determine $f \in X$
Step 1: Choose k output out of the n outputs, $\binom{n}{k}$ choices.
Step 2: Assign the inputs $1, 2, \ldots, k$ to the chosen k outputs, $k!$ choices.
Step 3: Assign the inputs $k+1, k+2, \ldots, n$ to the remaining $n-k$ outputs, $(n-k)!$
This shows that $|X| = \binom{n}{k} k!(n-k)!$ as desired. □

Another way of viewing X that can be more intuitive is to view it as the number of ways of arranging the numbers $\{1, 2, \ldots, n\}$ in a list of length n. After all, a bijection $f : [n] \to [n]$ is listing of the numbers $\{1, 2, \ldots, n\}$.

Example Let $n \geq 3$ be a natural number, then

$$\binom{n}{3} = \sum_{k=2}^{n-1} (k-1)(n-k)$$

Define X by

$$X = \{U \subseteq [n] : |U| = 3\}$$

Then $|X| = \binom{n}{3}$ by definition. Looking at the other side of the given expression, we will partition X in the following sets. For $2 \leq k \leq n-1$, define

$$A_k = \{U \in X : \text{ the middle element of } U \text{ is } k\}$$

Since the middle element of a subset of $[n]$ with three elements can only be between 2 and $n-1$, and there can't be two different values for the middle element in the same set, it is straightforward to show that $A_2, A_3, \ldots, A_{n-2}$ form a partition for X.

In order to determine an element U in A_k we
Step 1: Choose the smallest element in U, $k-1$ choices,
Step 2: Choose the largest element of U, $n-k$ choices. The MP shows that $|A_k| = (k-1)(n-k)$, for all k, $2 \leq k \leq n-1$. By the AP, we have

$$|X| = \sum_{k=2}^{n-1} |A_k| = \sum_{k=2}^{n-1} (k-1)(n-k)$$

By the double counting principle, we deduce that $\binom{n}{3} = \sum_{k=2}^{n-1}(k-1)(n-k)$.

Example Let n and k be natural numbers, then

$$\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$$

Define X by

$$X = \{U \subseteq [n+1] : |U| = k+1\}$$

Then $|X| = \binom{n+1}{k+1}$ by definition. We partition X as follows:

$$A = \{U \in X : n+1 \in U\} \qquad B = \{U \in X : n+1 \notin U\}$$

Since every set in X either has $n+1$ or does not have $n+1$, and not both, it is straightforward to show that A, B form a partition for X.
Since every element of A contains $n+1$, in order determined an element of A, we ought to choose the remaining k elements from $\{1, 2, \ldots, n\}$, therefore $|A| = \binom{n}{k}$.
Similarly, every element of B does not contain $n+1$, in order determined an element of B, we ought to choose all $k+1$ elements from $\{1, 2, \ldots, n\}$, therefore $|B| = \binom{n}{k+1}$.
By the AP

$$|X| = |A| + |B| = \binom{n}{k} + \binom{n}{k+1}$$

This shows that the given expression holds.

Example Let $m, n \in \mathbb{N}$. How many functions are there from $[m]$ to $[n]$?
In order to determine a function $f[m] \to [n]$ we
Step 1: Choose $f(1) \in [n]$, n choices,
Step 2: Choose $f(2) \in [n]$, n choices,
$\vdots$
Step m: Choose $f(m) \in [n]$, n choices.

We get a total of n^m functions.

Example Let S be the set of all functions from [9] to [4] that send exactly 3 inputs to 1. That is

$$S = \{f : [9] \to [4] : |f^{-1}[\{1\}]| = 3\}$$

Find $|S|$.
In order to determine a function $f \in S$ we
Step 1: Choose the 3 inputs that will go to 1, $\binom{9}{3}$ choices,
Step 2: Send the remaining 6 inputs to the remaining 3 outputs in any way, 3^6 choices.
Therefore, $|S| = \binom{9}{3}3^6$.

Combinatorics Exercises Solutions

1. (a) Given a group of 10 people, how many ways are there to choose a committee of at least 4 people and designate a president and vice-president from the committee? (The president and vice-president must be different people.)
   (b) Given a group of n people, how many ways are there to choose a committee of at least k people and designate a president and vice-president from the committee? (The president and vice-president must be different people.) We assume $k \geq 2$.
2. (a) How many functions $f : [4] \to [8]$ are there with $f[[4]] \subseteq \{2, 4, 6, 8\}$?
   (b) How many functions $f : [4] \to [8]$ are there with $f[[4]] = \{2, 4, 6, 8\}$?
3. Use a double-counting argument to show that for all $n, k \in \mathbb{N}$,
$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}.$$
4. Use a double-counting argument to show that for all $n, k \in \mathbb{N}$,
$$\sum_{i=k}^{n} \binom{i}{k} = \binom{n+1}{k+1}.$$
5. Let $n \in \mathbb{N}$. Prove that
$$1 \cdot 3 \cdot 5 \ldots \cdot (2n-3) \cdot (2n-1) = \frac{1}{n!}\binom{2}{2}\binom{4}{2}\binom{6}{2} \cdots \binom{2n-2}{2}\binom{2n}{2}$$
   in two ways, using algebra and using a double-counting argument.
6. Show that if $a, b, c \in \mathbb{N}$, then $a! \cdot b! \cdot c! \mid (a+b+c)!$.
7. (a) Find $|\{(a, b, c, d) \in \mathbb{N}^4 \mid a+b+c+d = 10\}|$, the number of 4-tuples of natural numbers whose sum is 10.
   (b) Find $|\{(a, b, c, d) \in (\mathbb{N} \setminus \{0\})^4 \mid a+b+c+d = 10\}|$, the number of 4-tuples of positive integers whose sum is 10.
8. Suppose Alice is dealt a hand of 5 cards from a standard deck of 52 cards. (The order of the cards in Alice's hand does not matter.)
   (a) If you know that Alice has 2 red cards and 3 black cards, how many possible hands can Alice have?
   (b) If you also know that exactly 1 of the 3 black cards is a spade, how many possible hands can Alice have?
   (c) If you also know that Alice's red cards have the same rank, how many possible hands can Alice have?

   We take the conditions to be cumulative, so in part (c), we also assume the conditions from (a) and (b).

Combinatorics Exercises Solutions

1. (a) Given a group of 10 people, how many ways are there to choose a committee of at least 4 people and designate a president and vice-president from the committee? (The president and vice-president must be different people.)

   (b) Given a group of n people, how many ways are there to choose a committee of at least k people and designate a president and vice-president from the committee? (The president and vice-president must be different people.) We assume $k \geq 2$.

   Solution.

   (a) Let X denote the set of possible committees of at least 4 people with a designated president and vice-president. For $4 \leq i \leq 10$, let X_i denote the set of committees from X with size i. Observe that $X_i \cap X_j = \emptyset$ for $i \neq j$ since there is no committee both with size i and with size j. Every committee in X must have size i with $i \geq 4$ and $i \leq 10$ since the committee can be no larger than the group of people, so the committee must be in some X_i. So $\{X_i \mid 4 \leq i \leq 10\}$ is a partition of X. By the addition principle,

   $$|X| = \sum_{i=4}^{10} |X_i|.$$

   Every element of X_i can be uniquely determined as follows:

   1. Choose a committee of size i from the group of 10 people. There are $\binom{10}{i}$ ways to do this.
   2. Choose a president from the committee. There are i ways to do this.
   3. Choose a vice-president from the committee, other than the president. There are $i - 1$ ways to do this.

   So by the multiplication principle,

   $$|X_i| = \binom{10}{i} i(i-1)$$

   for each $4 \leq i \leq 10$. Thus,

   $$|X| = \sum_{i=4}^{10} \binom{10}{i} i(i-1).$$

   (b) Let X denote the set of possible committees of at least k people with a designated president and vice-president. For $k \leq i \leq n$, define X_i to be the set of committees from X with size i. By the same argument as above, $\{X_k \mid k \leq i \leq n\}$ is a partition of X. By the addition principle,

   $$|X| = \sum_{i=k}^{n} |X_i|.$$

For $k \leq i \leq n$, we have $|X_i| = \binom{n}{i} i(i-1)$ by the same argument as in part (a). So

$$|X| = \sum_{i=k}^{n} \binom{n}{i} i(i-1).$$

2. (a) How many functions $f : [4] \to [8]$ are there with $f[[4]] \subseteq \{2, 4, 6, 8\}$?

   (b) How many functions $f : [4] \to [8]$ are there with $f[[4]] = \{2, 4, 6, 8\}$?

   Solution.

   (a) A function $f : [4] \to [8]$ with $f[[4]] \subseteq \{2, 4, 6, 8\}$ can be specified by the following:

   1. Choose $f(1)$ from $\{2, 4, 6, 8\}$. There are 4 choices.
   2. Choose $f(2)$ from $\{2, 4, 6, 8\}$. There are 4 choices.
   3. Choose $f(3)$ from $\{2, 4, 6, 8\}$. There are 4 choices.
   4. Choose $f(4)$ from $\{2, 4, 6, 8\}$. There are 4 choices.

   So by the multiplication principle, there are $4^4 = 256$ functions $f : [4] \to [8]$ with $f[[4]] \subseteq \{2, 4, 6, 8\}$.

   (b) Observe that if $f : [4] \to [8]$ with $f[[4]] = \{2, 4, 6, 8\}$, we must have $f(i) \neq f(j)$ if $i \neq j$. If not, we have

   $$|f[[4]]| \leq 3 < 4 = |\{2, 4, 6, 8\}|,$$

   contradicting $f[[4]] = \{2, 4, 6, 8\}$. So we can determine every $f : [4] \to [8]$ with $f[[4]] = \{2, 4, 6, 8\}$ via the following:

   1. Choose $f(1)$ from $\{2, 4, 6, 8\}$. There are 4 choices.
   2. Choose $f(2)$ from $\{2, 4, 6, 8\} \setminus \{f(1)\}$. There are 3 choices.
   3. Choose $f(3)$ from $\{2, 4, 6, 8\} \setminus \{f(1), f(2)\}$. There are 2 choices.
   4. Choose $f(4)$ from $\{2, 4, 6, 8\} \setminus \{f(1), f(2), f(3)\}$. There is 1 choice.

   By the multiplication principle, there are $4 \cdot 3 \cdot 2 \cdot 1 = 24$ functions $f : [4] \to [8]$ with $f[[4]] = \{2, 4, 6, 8\}$.

3. Use a double-counting argument to show that for all $n, k \in \mathbb{N}$,

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}.$$

   Solution. Let X be the set of all k-element subsets of $n+1$. We know $|X| = \binom{n+1}{k}$. Let Y be the set of all k-element subsets of $n+1$ that contain $n+1$, and let Z be the set of all k-element subsets of $n+1$ that do not contain $n+1$. Note that $\{Y, Z\}$ is a partition of X, so $|X| = |Y| + |Z|$ by the addition principle.

   Observe that every $A \in Y$ can be obtained by choosing a $(k-1)$-element subset $B \subseteq [n]$ and setting $A = B \cup \{n+1\}$. So $|Y| = \binom{n}{k-1}$. Every element of Z can be obtained by choosing a k-element subset of $[n]$, so $|Z| = \binom{n}{k}$. Therefore,

$$\binom{n+1}{k} = |X| = |Y| + |Z| = \binom{n}{k-1} + \binom{n}{k}.$$

4. Use a double-counting argument to show that for all $n, k \in \mathbb{N}$,

$$\sum_{i=k}^{n} \binom{i}{k} = \binom{n+1}{k+1}.$$

Solution. Let X be the set of $(k+1)$-element subsets of $n+1$. We know $|X| = \binom{n+1}{k+1}$. For $1 \leq j \leq n+1$, let $X_j = \{A \in X \mid \max(A) = j\}$. Note that $X_i \cap X_j = \emptyset$ if $i \neq j$. For any $A \in X$, A must have a maximum element $j \in [n+1]$ and so $A \in X_j$. So $\{X_j \mid j \in [n+1]\}$ is a partition of X. By the addition principle,

$$|X| = \sum_{j=1}^{n+1} |X_j|.$$

Let $j \in [n+1]$. Observe every element of X_j can be determined by choosing a k-element subset of $[j-1]$ and taking the union with $\{j\}$. So $|X_j| = \binom{j-1}{k}$. Then

$$|X| = \sum_{j=1}^{n+1} \binom{j-1}{k} = \sum_{i=0}^{n} \binom{i}{k}.$$

Observe that if $i < k$, then $\binom{i}{k} = 0$: every subset of an i-element set has at most i elements, so it cannot have k elements. Therefore,

$$\binom{n+1}{k+1} = |X| = \sum_{i=0}^{n} \binom{i}{k} = \sum_{i=k}^{n} \binom{i}{k}.$$

5. Let $n \in \mathbb{N}$. Prove that

$$1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-3) \cdot (2n-1) = \frac{1}{n!} \binom{2}{2} \binom{4}{2} \binom{6}{2} \cdots \binom{2n-2}{2} \binom{2n}{2}$$

in two ways, using algebra and using a double-counting argument.
Solution 1: Algebra. We know that for any $k \in \mathbb{N}$ with $k \geq 2$, $\binom{k}{2} = \frac{k!}{(k-2)!2!}$. Then

$$\binom{k}{2} = \frac{k!}{(k-2)!2!} = \frac{\prod_{i=1}^{k} i}{(\prod_{j=1}^{k-2} j) \cdot 2 \cdot 1} = \frac{k(k-1)}{2}.$$

So

$$\begin{aligned}
&\frac{1}{n!} \binom{2}{2} \binom{4}{2} \binom{6}{2} \cdots \binom{2n-2}{2} \binom{2n}{2} \\
=&\frac{1}{n!} \cdot \frac{2 \cdot 1}{2} \cdot \frac{4 \cdot 3}{2} \cdot \frac{6 \cdot 5}{2} \cdot \ldots \cdot \frac{(2n-2)(2n-3)}{2} \cdot \frac{2n(2n-1)}{2} \\
=&\frac{1}{n!} \cdot (1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)) \cdot \frac{2 \cdot 4 \cdot 6 \cdot \ldots \cdot 2n}{2^n} \\
=&\frac{1}{n!} \cdot (1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)) \cdot (1 \cdot 2 \cdot 3 \cdot \ldots \cdot n) \\
=&1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1).
\end{aligned}$$

Solution 2: Double-counting. We will count the number of ways to divide $[2n]$ into n (unordered) pairs such that every number appears in exactly one pair. Let k denote this number. Note that a procedure to split $[2n]$ into n pairs with every number appearing in exactly one pair is:

1. Choose an element $a_1 \in [2n] \setminus \{1\}$ and let $\{1, a_1\}$ be a pair.
2. Let b_2 denote the least element of $[2n] \setminus \{1, a_1\}$. Choose an element $a_2 \in [2n] \setminus \{1, a_1, b_2\}$, and let $\{b_2, a_2\}$ be a pair.
3. Repeat this process until we have n pairs.

There are $2n-1$ choices for a_1, $2n-3$ choices for a_2, $2n-5$ choices for a_3, …, and 1 choice for a_n, so $k = 1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-3) \cdot (2n-1)$.

We now consider choosing pairs one at a time. Consider the following procedure:

1. Choose a pair $\{a_1, b_1\}$ from $[2n]$.
2. Choose a pair $\{a_2, b_2\}$ from $[2n] \setminus \{a_1, b_1\}$.
3. Choose a pair $\{a_3, b_3\}$ from $[2n] \setminus \{a_1, b_1, a_2, b_2\}$.
4. Repeat until we have n pairs.

Note that the number of ways to run this procedure is $n! \cdot k$ since n pairs can be picked in any of $n!$ possible orders. So $n! \cdot k = \binom{2n}{2}\binom{2n-2}{2}\cdots\binom{4}{2}\binom{2}{2}$, hence $k = \frac{1}{n!}\binom{2}{2}\binom{4}{2}\cdots\binom{2n-2}{2}\binom{2n}{2}$.

Thus, we conclude $1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-3) \cdot (2n-1) = \frac{1}{n!}\binom{2}{2}\binom{4}{2}\cdots\binom{2n-2}{2}\binom{2n}{2}$.

6. Show that if $a, b, c \in \mathbb{N}$, then $a! \cdot b! \cdot c! \mid (a+b+c)!$.

Solution. Let X be the set of tuples (A, B, C) such that $\{A, B, C\}$ is a partition of $[a+b+c]$ where $|A| = a$, $|B| = b$, and $|C| = c$. Consider the following procedure:

(a) List the elements of $[a+b+c]$ in some order.

(b) Let A be the first a elements, B the next b elements, and C the next c elements.

There are $(a+b+c)!$ ways to carry out this procedure. Note that this procedure produces every element of X exactly $a! \cdot b! \cdot c!$ times: there are $a!$ ways to reorder elements of A, $b!$ ways to reorder elements of B, and $c!$ ways to reorder elements of C. Then $|X| \cdot (a! \cdot b! \cdot c!) = (a+b+c)!$, and since $|X|$ is an integer, we have $a! \cdot b! \cdot c! \mid (a+b+c)!$.

7. (a) Find $|\{(a, b, c, d) \in \mathbb{N}^4 \mid a+b+c+d = 10\}|$, the number of 4-tuples of natural numbers whose sum is 10.

(b) Find $|\{(a, b, c, d) \in (\mathbb{N} \setminus \{0\})^4 \mid a+b+c+d = 10\}|$, the number of 4-tuples of positive integers whose sum is 10.

Solution.

(a) We can produce a tuple in this set as follows:

1. Put 10 balls and 3 dividers in a line in any order.
2. Let a be the number of balls to the left of the leftmost divider.
3. Let b be the number of balls between the leftmost divider and the middle divider.
4. Let c be the number of balls between the middle divider and the rightmost

divider.

5. Let d be the number of balls to the right of the rightmost divider.

Note that there are $\binom{13}{3}$ ways to order the balls and dividers: there are 13 objects we need to order, and we can pick 3 positions for the dividers and let the balls take the remaining 10 positions. So $|\{(a, b, c, d) \in \mathbb{N}^4 \mid a + b + c + d = 10\}| = \binom{13}{3}$.

(b) We can produce every tuple in this set by choosing a tuple $(a', b', c', d') \in \mathbb{N}^4$ with $a' + b' + c' + d' = 6$ and setting $a = a' + 1$, $b = b' + 1$, $c = c' + 1$, $d = d' + 1$. A similar argument to part (a) gives that there are $\binom{9}{3}$ ways to choose (a', b', c', d'), so $|\{(a, b, c, d) \in (\mathbb{N} \setminus \{0\})^4 \mid a + b + c + d = 10\}| = \binom{9}{3}$.

8. Suppose Alice is dealt a hand of 5 cards from a standard deck of 52 cards. (The order of the cards in Alice's hand does not matter.)

(a) If you know that Alice has 2 red cards and 3 black cards, how many possible hands can Alice have?

(b) If you also know that exactly 1 of the 3 black cards is a spade, how many possible hands can Alice have?

(c) If you also know that Alice's red cards have the same rank, how many possible hands can Alice have?

We take the conditions to be cumulative, so in part (c), we also assume the conditions from (a) and (b).

Solution.

(a) There are 26 red cards and 26 black cards in the deck. Any hand with 2 red cards and 3 black cards can be formed by choosing 2 of the 26 red cards and 3 of the 26 black cards, so there are $\binom{26}{2}\binom{26}{3}$ possible hands.

(b) We know that Alice has 2 red cards, 1 spade, and 2 clubs. So any possible hand can be formed by choosing 2 of the 26 red cards, 1 of the 13 spades, and 2 of the 13 clubs. So there are $\binom{26}{2}\binom{13}{1}\binom{13}{2}$ possible hands.

(c) For the red cards in Alice's hand, Alice must have 1 heart and 1 diamond of the same rank, and there are 13 possible ranks. So any possible hand is formed by choosing 1 of the 13 hearts, adding in the diamond of the same rank, choosing 1 of the 13 spades, and choosing 2 of the 13 clubs. There are $\binom{13}{1}\binom{13}{1}\binom{13}{2}$ possible hands.